

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Theory Examination October 2017

PH222 PHYTOCHEMISTRY-I

Date: 05/10/2017, Thursday

Time: 1:30 pm to 4:30 pm

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q.1 Attempt all questions. Each question is of one mark.

20

- 1 One of the following is source of rutin
A. Buckwheat
B. Dioscorea
C. Digitalis
D. Aloe
- 2 Which drug has laxative property?
A. Digitoxin
B. Sennoside
C. Glycyrrhizin
D. All of above
- 3 _____ is used in treatment of cancer
A. Vincristine
B. Rutin
C. Atropine
D. Quinine
- 4 _____ is a lysergic acid derivative
A. Quinine
B. Morphine
C. Ergotamine
D. Ephedrine
- 5 One of the following is proto alkaloid.
A. Reserpine
B. Quinine
C. Ephedrine
D. Tubocurarine
- 6 Glycyrrhithinic acid is example of
A. Saponin glycoside
B. Cardiac glycoside
C. Indole alkaloid
D. Anthraquinone glycoside

- 7 Morphine is obtained from _____
A. Vinca spp.
B. Cinchona spp.
C. Papaver spp.
D. Claviceps spp.
- 8 Vitamin P is also known as _____
A. Permeability Factor
B. Rutin
C. A and B both
D. Vincristine
- 9 Podophyllotoxin is an example
A. Lignan
B. Flavonoid
C. Terpene
D. Coumarin
- 10 Morphine is obtained from _____ of Papaver species
A. Leaves
B. Stem
C. Root
D. Exudate
- 11 Application of acid in glycoside isolation is required to _____
A. Hydrolyze aglycone
B. Hydrolyze Glycone
C. Hydrolyze glycosidic Linkage
D. Solubilize glycoside
- 12 Which of the following is biosynthesized from Tyrosine
A. Morphine
B. Ergometrine
C. Diosgenin
D. Sennoside
- 13 _____ is used as central analgesic.
A. Vincristine
B. Caffeine
C. Morphine
D. Podophyllotoxin
- 14 Glycosides are _____
A. Primary products
B. Secondary metabolites
C. Always solid in state
D. White in color
- 15 _____ has antihypertensive property,
A. Quinine
B. Quinidine
C. Reserpine
D. Aloin
- 16 To liberate free alkaloid as base in isolation, _____ is added
A. Potassium hydroxide
B. Sulphuric acid

- C. Hydrochloric acid
- D. Benzene
- 17 Pathway involved in biosynthesis of ephedrine is
 - A. Acetate melonate pathway
 - B. Acetate mevalonate pathway
 - C. Acetate glyconate pathway
 - D. Shikimic acid pathway
- 18 According to Isoprene rule, two isoprene units are joined in which manner?
 - A. Head to head
 - B. Tail to tail
 - C. Head to tail
 - D. None
- 19 Find the odd thing
 - A. Amino acids
 - B. Terpenoids
 - C. Fatty acids
 - D. Carbohydrates
- 20 Quinine is _____
 - A. Anti malarial
 - B. anti viral
 - C. Anti arrhythmic
 - D. Anti hypertensive

SECTION – II

Q.2 Attempt any TWO of the following:

- A Define primary and secondary metabolites of plants giving examples. Describe its inter-relationship. 05
- B Prepare a note on Acetate melonate pathway. 05
- C Write source, biosynthesis and uses of Rutin. 05

Q.3 Attempt any TWO of the following:

- A Describe in detail about source, biosynthesis and uses of caffeine. 05
- B Answer the following: 05
 - 1) Why lead acetate and sulfuric acid are added for isolation of caffeine?
 - 2) Why tobacco leaf powder are moistened with KOH?
- C How radio isotope tracer technique is useful in biosynthesis? 05

SECTION – III

Q.4 Attempt any FOUR of the following:

- A Write source and describe steps involved in isolation of Nicotine with justification. 05
- B Describe important steps involved in generation of tereponoidal compounds from acetate as precursor. 05
- C What is stereo isomerism? Discuss stereoisomerism of natural compounds with suitable example 05
- D Discuss in detail about biosynthesis of Diosgenin. 05
- E Write the source of the followings: 05
1) Morphine 2) Aloin 3) Ephedrine 4) Nicotine 5) Podophyllotoxin
- F Explain biosynthesis of Glycyrrhizin. 05

Q.5 Attempt any FOUR of the following:

- A Prepare a note on inter relationship between *Digitalis purpurea* and *Digitalis lanata* glycosides. 05
 - B Explain chemistry, isolation and estimation of glycyrrhizin. 05
 - C Write the uses of followings: 05
(1) Reserpine (2) Rutin (3) Ephedrine (4) Morphine (5) Aloin
 - D Write in detail about source and medicinal utilization of Vinca alkaloid. 05
 - E Write in brief about biosynthetic production of Aloin 05
 - F Describe pharmacological importance and write utilization of Ergometrine. 05
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